

Gaps between Turán densities for r -graphs on $r + 1$ vertices

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Abstract

For $1 \leq t \leq r + 1$, let $H_t^{(r)}$ denote the r -graph with $r + 1$ vertices and t edges. We determine the complete order of their Turán densities: for every $r \geq 2$, $\pi(H_1^{(r)}) = \pi(H_2^{(r)}) = 0 < \pi(H_3^{(r)}) < \dots < \pi(H_{r+1}^{(r)})$. We also give explicit lower bounds for every positive adjacent gap, including estimates that are uniform over the interior range. The proof combines refined blowup constructions with sparse packings, while the comparison between three and four edges requires a separate graph-coloring argument. The resulting gap bounds yield exact anti-Ramsey formulas for up to a constant multiple of $\sqrt{n}$ disjoint copies of $H_t^{(r)}$. We further prove interlacing inequalities across uniformities, which give additional lower bounds for adjacent gaps along infinite subsequences and within finite windows.

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1 Introduction

For an integer $r \geq 2$, an r -uniform hypergraph, or briefly an r -graph, G on vertex set $V(G)$ is a family $G \subseteq \binom{V(G)}{r}$. We identify a hypergraph with its edge set, and hence write $|H|$ for the number of edges of H . We also write $v(H) := |V(H)|$. Given r -graphs F and G , we say that G is F -free if it contains no copy of F as a subgraph. Copies are not required to be induced. We also write $[n] := \{1, \dots, n\}$ and use $O_r(\cdot)$ when the implicit constant may depend on r . The Turán number $\text{ex}(n, F)$ is the maximum number of edges in an F -free r -graph on n vertices. The Turán density of F is $\pi(F) := \lim_{n \rightarrow \infty} \text{ex}(n, F) / \binom{n}{r}$. The limit exists by the standard averaging argument of Katona, Nemetz, and Simonovits [16]. For general background on hypergraph Turán problems, we refer to the survey of Keevash [17]. For a finite family $\mathcal{F}$ of r -graphs, the quantities $\text{ex}(n, \mathcal{F})$ and $\pi(\mathcal{F})$ are defined analogously by forbidding every member of $\mathcal{F}$.

Let $K_s^{(r)}$ be the complete r -graph on s vertices, and let $K_s^{(r)-}$ be obtained from it by deleting one edge. In particular, $K_{r+1}^{(r)}$ is the r -uniform simplex. For $1 \leq t \leq r + 1$, let $H_t^{(r)}$ denote the r -graph with $r + 1$ vertices and t edges; it is unique up to isomorphism. For an $(r + 1)$ -set S , we call the sets $S \setminus \{x\}$, $x \in S$, its facets. Put $p_{r,t} := \pi(H_t^{(r)})$. Equivalently, an r -graph is $H_t^{(r)}$ -free if and only if every $(r + 1)$ -set of vertices spans at most $t - 1$ edges. Thus $H_r^{(r)} = K_{r+1}^{(r)-}$ and $H_{r+1}^{(r)} = K_{r+1}^{(r)}$. Since $H_t^{(r)} \subseteq H_{t+1}^{(r)}$, the sequence $p_{r,1}, p_{r,2}, \dots, p_{r,r+1}$ is nondecreasing. Moreover, $p_{r,1} = p_{r,2} = 0$, so the first possible strict comparison is $p_{r,2} < p_{r,3}$.

The Turán problems for this family are instances of the local-density problems introduced by Brown, Erdős, and Sós [4]. Moreover, $H_t^{(r)}$ is the r -daisy $D_r(t, t - 1)$ in the terminology of Bollobás, Leader, and Malvenuto [3]. Markström and Thomassen [23, Corollary 3.9] proved the general bound $\pi(H_t^{(r)}) \leq (t - 2)/r$ for $t \geq 3$. For every fixed $t \geq 3$, Sidorenko [28, Corollary 3.2] proved $p_{r,t} = \Omega(r^{-(1+1/(t-2))})$ as $r \rightarrow \infty$. For the first nontrivial density, Clemen [5, Theorem 1.2] subsequently proved $p_{r,3} = \Omega(r^{-2}\sqrt{\log r})$ as $r \rightarrow \infty$. More recently, Yang and Zhu [30] proved $p_{r,3} = r^{-1+o(1)}$ and obtained uniform quantitative estimates for the individual values $p_{r,t}$ throughout $3 \leq t \leq r + 1$, whereas our main result determines their fixed-uniformity ordering.

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The anti-Ramsey problem was introduced by Erdős, Simonovits, and Sós [7]. For a positive integer s , let sF be the r -graph consisting of s vertex-disjoint copies of an r -graph F . A copy of sF in an edge-colored complete r -graph is *rainbow* if all of its edges receive distinct colors. We use the convention that the *anti-Ramsey number* $\text{ar}(n, sF)$ is the least integer N such that every edge-coloring of $K_n^{(r)}$ using exactly N colors contains a rainbow copy of sF . If $\mathcal{F}_-$ is the family of all one-edge deletions of F , then Erdős, Simonovits, and Sós proved $\text{ex}(n, \mathcal{F}_-) + 2 \leq \text{ar}(n, F) \leq \text{ex}(n, \mathcal{F}_-) + o(n^r)$. The density results for partial hypergraph tilings obtained by Hou, Li, Liu, Yuan, and Zhang [15], including their rainbow analogue, are directly relevant here.

Our direct motivation comes from the rainbow partial hypergraph tilings studied by Deng, Hou, Liu, and Yang. Combining Proposition 1.6(i) with Theorem 1.5 in their paper, Deng, Hou, Liu, and Yang [6] proved that $\pi(\mathcal{F}_-) < \pi(F)$ implies $\text{ar}(n, 2F) = \text{ex}(n, F) + 2$ for all sufficiently large n . Since all one-edge deletions of $K_\ell^{(r)}$ are isomorphic, they asked whether $\pi(K_\ell^{(r)-}) < \pi(K_\ell^{(r)})$ for all $\ell > r \geq 3$ [6, Problem 6.2].

The same inequality also appears as the endpoint of a question of Liu, Schülke, Wang, Yang, and Zhang. They developed a general method for separating hypergraph Turán densities and proved [18, Corollary 1.6] that $\pi(H_t^{(r)}) < \pi(H_{t+2}^{(r)})$ for $1 \leq t \leq r-1$. They asked [18, Problem 4.2] whether $\pi(H_t^{(r)}) < \pi(H_{t+1}^{(r)})$ for $3 \leq t \leq r$. We settle all of these adjacent comparisons, together with the elementary first nontrivial step.

Theorem 1.1. *For every $r \geq 2$, we have $p_{r,1} = p_{r,2} = 0 < p_{r,3} < \dots < p_{r,r+1}$.*

The endpoint argument is also quantitative: it gives $p_{r,r+1} - p_{r,r} \geq (2 - o(1))r^{-6}$ as $r \rightarrow \infty$; the full explicit estimate is stated in Theorem 3.1.

The arguments give quantitative lower bounds for the remaining gaps as well. We record the uniform estimate for the interior gaps and an explicit bound for the critical gap in one statement. For an integer $k \geq 1$, let $R_k(3)$ denote the least N such that every coloring $\varphi: K_N \rightarrow [k]$ contains a monochromatic triangle.

Theorem 1.2. *The following estimates hold.*

(i) *Uniformly over $4 \leq t \leq r-1$ as $r \rightarrow \infty$,*

$$p_{r,t+1} - p_{r,t} \geq (2 - o(1)) \frac{(r-t+2)^2}{r^5(t-1)^2(t-2)}.$$

(ii) *For $r \geq 3$, put $M_r := 200R_{r-1}(3)r(r+1) + 1$. Then*

$$p_{r,4} - p_{r,3} \geq \frac{r-1}{800R_{r-1}(3)r^2(r+1)} M_r^{-2M_r} > 0.$$

Part (i) of Theorem 1.2 has the following useful asymptotic forms:

$$\begin{aligned} p_{r,t+1} - p_{r,t} &\geq \left(\frac{2}{(t-1)^2(t-2)} - o(1) \right) r^{-3} && \text{if } t \geq 4 \text{ is fixed,} \\ p_{r,t+1} - p_{r,t} &\geq \left(\frac{2(1-\beta)^2}{\beta^3} - o(1) \right) r^{-6} && \text{if } t/r \rightarrow \beta \in (0, 1), \\ p_{r,r-\ell+1} - p_{r,r-\ell} &\geq \frac{2(\ell+2)^2 - o(1)}{r^8} && \text{if } r-t = \ell \geq 1 \text{ is fixed.} \end{aligned}$$

Using the classical bound $R_{r-1}(3) \leq 3(r-1)!$ [13, Theorem 6], we can also remove the Ramsey number from part (ii) of Theorem 1.2. Namely, with $\widehat{M}_r := 600r(r+1)(r-1)! + 1$, we obtain

$$p_{r,4} - p_{r,3} \geq \frac{\widehat{M}_r^{-2\widehat{M}_r}}{2400r^2(r+1)(r-2)!}.$$

Together with the elementary comparison $p_{r,2} < p_{r,3}$ proved below, the estimates in Theorems 3.1 and 4.1 and part (ii) of Theorem 1.2 establish Theorem 1.1.

We next place the strict chain in a broader context and record some related consequences.

From a broader spectrum viewpoint, Liu and Pikhurko [19, Theorem 1.1] proved that, for every $r \geq 3$, the Turán densities of possibly infinite forbidden r -graph families contain nondegenerate intervals, including one with right endpoint 1. Here we study the more rigid sequence of consecutive densities arising from one nested family of single forbidden r -graphs.

For $r = 3$, the iterated construction of Frankl and Füredi [10] gives $p_{3,3} \geq 2/7$. Baber and Talbot [2] proved $p_{3,3} \leq 0.2871$, and Falgas-Ravry and Vaughan [9] improved this to $p_{3,3} \leq 0.286889$. Turán's construction [29] gives $p_{3,4} \geq 5/9$, while Baber [1] proved $p_{3,4} \leq 0.5615$; the exact value remains open. Razborov [27] obtained the matching upper bound $5/9$ under the additional restriction that no four vertices span exactly one edge.

For $r = 4$, Gunderson and Semeraro [14] proved $p_{4,3} = 1/4$. The bound of Markström and Thomassen [23] gives $p_{4,4} \leq 1/2$, while Giraud's construction [11] and Markström's upper bound [22] give $11/16 \leq p_{4,5} \leq 1753/2380$. The elementary endpoint arguments for $r = 2$ and $r = 3$ below are included for completeness.

There is an equivalent covering formulation. Let $T_j(n; r+1, r)$ be the minimum size of a family $\mathcal{D} \subseteq \binom{[n]}{r}$ such that every $(r+1)$ -set contains at least j members of $\mathcal{D}$. The case $j = 1$ is the usual *Turán-system number* $T(n, r+1, r)$, so that $T_1(n; r+1, r) = T(n, r+1, r)$; see Pikhurko [26]. For recent upper bounds on $T(n, s, r)$ in the intermediate regime $1 \ll s - r = O(r/\log r)$, see Liu and Pikhurko [20]. For $1 \leq j \leq r+1$, taking complements and then passing to the limit give

$$T_j(n; r+1, r) = \binom{n}{r} - \text{ex}(n, H_{r+2-j}^{(r)}) \quad \text{and} \quad \tau_j^{(r)} := \lim_{n \rightarrow \infty} \frac{T_j(n; r+1, r)}{\binom{n}{r}} = 1 - p_{r, r+2-j}.$$

Equivalently, $p_{r,t} = 1 - \tau_{r-t+2}^{(r)}$ for $1 \leq t \leq r+1$. Hence, for $1 \leq t \leq r$, $p_{r,t+1} - p_{r,t} = \tau_{r-t+2}^{(r)} - \tau_{r-t+1}^{(r)}$. In particular, $\tau_2^{(r)} - \tau_1^{(r)} = p_{r,r+1} - p_{r,r}$. Thus Theorem 1.1 is equivalent to $\tau_1^{(r)} < \tau_2^{(r)} < \dots < \tau_r^{(r)} = \tau_{r+1}^{(r)} = 1$ for every $r \geq 2$.

This chain compares covering densities at one fixed uniformity. There is also an ordered-deletion comparison that preserves normalized density across different uniformities. For $r \geq 3$, $1 \leq k \leq r-2$, and $1 \leq t \leq r-k+1$, the construction in Section 6 yields $p_{r,t} \leq p_{r-k,t} \leq p_{r,t+k}$. The corresponding covering inequalities are given in Corollary 6.2. The one-step interlacing also decomposes every adjacent gap into two nonnegative terms. Combining Proposition 6.3 with Sidorenko's fixed-level lower bounds gives quantitative subsequence and finite-window estimates; see Corollary 6.4.

We conclude the introduction with an outline of the proofs. Let G be a finite $K_{r+1}^{(r)-}$ -free r -graph on $[n]$. An ordinary equal blowup of G keeps only transverse edges and has limiting density

$$\frac{r!|G|}{n^r} = \frac{r! \binom{n}{r}}{n^r} \frac{|G|}{\binom{n}{r}} = \left(1 - \frac{\binom{r}{2}}{n} + O_r(n^{-2})\right) \frac{|G|}{\binom{n}{r}}.$$

The first-order loss is too large to obtain a strict separation by adding only a sparse family of missing template edges.

To recover this loss, for every template vertex i we choose a random root $\rho_i \neq i$ and use the pair-link of $\{i, \rho_i\}$ to define blowup edges whose label multiplicities are $2, 1, \dots, 1$. The expected contribution of these pair-links cancels the first-order density loss from repeated labels. We then add a weighted packing of missing template edges, chosen so that no two selected r -sets share $r-1$ vertices; equivalently, the packing is independent in the relevant Johnson graph. A modular n -coloring retains a $1/n$ fraction of the total positive weight. The only possible conflict between the two layers corresponds to an $(r+1)$ -set on which G has exactly $r-1$ edges. An exact expectation identity counts these conflicts, and deleting one pair-link edge for every selected conflict makes the construction simplex-free. After this repair, the packing gain has order n^{-1} , while the remaining blowup loss has order n^{-2} .

For the interior steps, independent roots spread the repair loss over too many missing-facet pairs. We instead choose h uniformly from $[n]$, set every non-hub root equal to h , and give the hub itself one additional random root. This correlation replaces a quadratic missing-facet penalty by a linear one. A modular n -coloring of the Johnson graph then retains a $1/n$ fraction of the positive packing weight. Finally, a convex combination of two finite templates with densities immediately below and above $p_{r,t}$ calibrates the affine density inequality exactly.

The pair-link layer does not preserve $H_4^{(r)}$ -freeness at the three-edge source level because the multiplicity pattern $2, 2, 1^{r-3}$ can create four pair-link facets. For this critical step we instead superimpose a second

isomorphic $H_3^{(r)}$ -free template. Two candidate edges are declared adjacent precisely when inserting both would create four edges on one $(r+1)$ -set. The resulting conflict graph has maximum degree $O(n)$ and clique number bounded in terms of r . Molloy's coloring theorem then supplies an independent family with an additional $\log n / \log \log n$ factor, which eventually dominates the first-order loss of the transverse blowup.

We collect finite monotonicity, modular Johnson packing, and second-order product estimates in [Section 2](#). The endpoint construction, its repair, the second-order blowup, and the resulting density estimate are developed together in [Section 3](#). The correlated hub-root construction for the interior gaps appears in [Section 4](#), and the conflict-graph argument based on a second template is given in [Section 5](#). Ordered lifts, interlacing, tail sums, and consequences of Sidorenko's lower bounds are treated in [Section 6](#).

2 Preliminaries

This section settles the elementary gap $p_{r,2} < p_{r,3}$ and records finite monotonicity for selecting dense templates, a modular coloring for weighted Johnson packings, and a second-order product estimate.

If G is an r -graph and $S \subseteq V(G)$, then $G[S]$ denotes the *induced r -graph* on S .

Lemma 2.1 ([8, Proposition 1]). *For every r -graph F , the sequence $\text{ex}(n, F) / \binom{n}{r}$ is nonincreasing for $n \geq v(F)$. In particular, $\text{ex}(n, F) / \binom{n}{r} \geq \pi(F)$ throughout this range.*

Lemma 2.2 ([10, 21]). *For every $r \geq 2$, we have $p_{r,2} = 0 < p_{r,3}$. More precisely, $p_{2,3} = 1/2$, while $p_{r,3} \geq 2^{1-r}$ for $r \geq 3$.*

Indeed, in an $H_2^{(r)}$ -free r -graph every $(r-1)$ -set has codegree at most one, so $r|G| \leq \binom{n}{r-1}$ and the normalized density tends to zero.

The *Johnson graph* $J(n, r)$ has vertex set $\binom{[n]}{r}$, with two r -sets adjacent exactly when their intersection has size $r-1$. We call a family of r -sets *Johnson-independent* if it is independent in $J(n, r)$. The following weighted form of the standard modular coloring will be used in both pair-link constructions.

Lemma 2.3 ([12, Theorem 1]). *Let $n \geq r \geq 2$ be integers, let $\mathcal{W} \subseteq \binom{[n]}{r}$, and let $w: \mathcal{W} \rightarrow \mathbb{R}$. There is a family $D \subseteq \mathcal{W}$, containing only positive-weight members, such that distinct members of D intersect in at most $r-2$ vertices and $\sum_{L \in D} w(L) \geq n^{-1} \sum_{L \in \mathcal{W}} w(L)$.*

To obtain the stated form, apply the residue-sum coloring to the positive-weight members of $\mathcal{W}$ and choose a heaviest color class; if there are no positive-weight members, take $D = \emptyset$.

We also need the following uniform second-order estimate. For integers $r \geq 2$ and $n > r$, put $s := \binom{r}{2}$ and $\kappa_r := r(r-1)(r-2)(3r+7)/24$. Define $A_{r,n} := \prod_{u=0}^{r-1} (1 - u/n)$ and $B_{r,n} := A_{r,n}(1 + s/(n-1))$.

Lemma 2.4. *For all integers $r \geq 2$ and $n > r$, we have $B_{r,n} \leq 1$ and $A_{r,n} \geq 1 - s/n$. Moreover, if $n \geq r(r-1)(r-2)/(4(r+2))$, then $B_{r,n} \geq 1 - \kappa_r/n^2$.*

Proof. First,

$$A_{r,n}^{-1} = \prod_{u=1}^{r-1} \left(1 + \frac{u}{n-u}\right) \geq 1 + \sum_{u=1}^{r-1} \frac{u}{n-u} \geq 1 + \frac{s}{n-1},$$

which proves $B_{r,n} \leq 1$. The elementary product inequality $\prod_u (1 - x_u) \geq 1 - \sum_u x_u$ for $0 \leq x_u \leq 1$ gives the lower bound for $A_{r,n}$.

For the second-order lower bound on $B_{r,n}$, the second and third elementary symmetric sums in $1, 2, \dots, r-1$ are $e_2 := r(r-1)(r-2)(3r-1)/24$ and $e_3 := r^2(r-1)^2(r-2)(r-3)/48$. Bonferroni's inequality yields $A_{r,n} \geq 1 - s/n + e_2/n^2 - e_3/n^3$. Using $\kappa_r = s^2 - s - e_2$, a direct simplification gives

$$\begin{aligned} & \left(1 - \frac{s}{n} + \frac{e_2}{n^2} - \frac{e_3}{n^3}\right) \left(1 + \frac{s}{n-1}\right) - \left(1 - \frac{\kappa_r}{n^2}\right) \\ &= \frac{r(r-1)(r-2)(r-3)(r+1)(r+2)}{24n^2(n-1)} \left(1 - \frac{r(r-1)(r-2)}{4(r+2)n}\right), \end{aligned}$$

which is nonnegative under the hypothesis in the second assertion. $\square$

For later use, note that for all positive γ , K , and n ,

$$\frac{\gamma}{n} - \frac{K}{n^2} = \frac{\gamma^2}{4K} \left[1 - \left(\frac{n - 2K/\gamma}{n} \right)^2 \right]. \quad (2.1)$$

3 The endpoint gap

The endpoint comparison in [Theorem 1.1](#) admits an explicit quantitative form. For every $r \geq 4$, define

$$n_r^{\text{end}} := \left\lceil r(r-1) + \frac{(r-1)^2(r-2)^2(3r+7)}{12} \right\rceil, \quad \Delta_r^{\text{end}} := \frac{6(1 - (n_r^{\text{end}})^{-2})}{(r-1)^2((r-1)(r-2)^2(3r+7) + 12r)}.$$

We will in fact prove the following quantitative estimate.

Theorem 3.1. *We have $p_{2,3} - p_{2,2} = 1/2$ and $p_{3,4} - p_{3,3} \geq 2/9$. For every $r \geq 4$, we have $p_{r,r+1} - p_{r,r} \geq \Delta_r^{\text{end}} > 0$. Consequently, $p_{r,r+1} - p_{r,r} \geq (2 - o(1))r^{-6}$ as $r \rightarrow \infty$.*

For every $r \geq 3$, this settles the case $\ell = r + 1$ of the question of Deng et al. [\[6, Problem 6.2\]](#); the theorem also covers $r = 2$.

The general separation criterion of Liu, Schülke, Wang, Yang, and Zhang [\[18, Theorem 1.4\]](#) does not cover this endpoint. For $r \geq 3$, neither of its conditions separates $F = K_{r+1}^{(r)-}$ from $F' = K_{r+1}^{(r)-}$. The numerical condition is tight rather than strict: $\text{ex}(r+1, F') + \prod_{i=0}^{r-1} \lfloor (r+1+i)/r \rfloor = (r-1) + 2 = r+1 = |F|$. The structural condition is also inconclusive. Take one part of size two and $r-1$ singleton parts. Every edge of F meets the first part, whereas deleting any singleton leaves only r vertices and hence no copy of F' .

We first augment a finite $K_{r+1}^{(r)-}$ -free template by combining a weighted Johnson packing with pair-links. A local repair makes the two layers compatible, after which a second-order blowup converts their total weight into a simplex-free density. We then calibrate the finite inequality at the exact density $p_{r,r}$.

We begin with the packing and repair step. Let $r \geq 4$ and $n > r$, and let G be a $K_{r+1}^{(r)-}$ -free r -graph on $[n]$. Put $\mathcal{C} := \binom{[n]}{r} \setminus G$, and let $\mathcal{T}$ be the family of sets $S \in \binom{[n]}{r+1}$ with $|G[S]| = r-1$. Thus every member of $\mathcal{T}$ has exactly two missing facets.

The following proposition combines the packing, pair-link, and repair steps.

Proposition 3.2. *There exist a family $D \subseteq \mathcal{C}$ and $(r-2)$ -graphs $B'_i \subseteq \binom{[n] \setminus \{i\}}{r-2}$ for $i \in [n]$ such that the following hold.*

- (i) *If $L, L' \in D$ are distinct, then $|L \cap L'| \leq r-2$.*
- (ii) *For every $L \in G \cup D$ and every $i \in L$, we have $\binom{L \setminus \{i\}}{r-2} \not\subseteq B'_i$.*
- (iii) *We have*

$$|G| + |D| + \frac{1}{2} \sum_{i=1}^n |B'_i| \geq \left(1 + \frac{s}{n-1}\right) |G| + \frac{1}{n} \left(|\mathcal{C}| - \frac{|\mathcal{T}|}{n-1} \right).$$

Proof. Random roots and pair-links.

For each $i \in [n]$, choose independently and uniformly a root $\rho_i \in [n] \setminus \{i\}$. Define the $(r-2)$ -graph

$$B_i := \left\{ U \in \binom{[n] \setminus \{i, \rho_i\}}{r-2} : U \cup \{i, \rho_i\} \in G \right\}.$$

Thus B_i is the *pair-link* of $\{i, \rho_i\}$ in G . Since each ρ_i is uniform in $[n] \setminus \{i\}$, double-counting ordered pairs of vertices in edges gives

$$\mathbb{E} \sum_{i=1}^n |B_i| = \frac{r(r-1)}{n-1} |G| = \frac{2s}{n-1} |G|. \quad (3.1)$$

Bad incidences.

For $L \in \mathcal{C}$ and $i \in L$, call (L, i) a *bad incidence* if $\binom{L \setminus \{i\}}{r-2} \subseteq B_i$. Let $c(L)$ be the number of bad incidences involving L , and assign to L the *weight* $w(L) := 1 - c(L)/2$. The unit term is the gain from inserting L

into the transverse template. Each bad incidence may require the deletion of one pair-link edge, which has weight $1/2$ in the final blowup count.

The following exact identity is the key local calculation.

Bad-incidence identity. We have

$$\mathbb{E} \sum_{L \in \mathcal{C}} c(L) = \frac{2|\mathcal{T}|}{n-1} \quad \text{and} \quad \mathbb{E} \sum_{L \in \mathcal{C}} w(L) = |\mathcal{C}| - \frac{|\mathcal{T}|}{n-1}.$$

To prove these identities, suppose that (L, i) is a bad incidence and put $a := \rho_i$. First, $a \notin L$. Indeed, if $a \in L \setminus \{i\}$, then some $(r-2)$ -subset of $L \setminus \{i\}$ contains a , whereas every member of B_i is disjoint from $\{i, a\}$.

Set $S := L \cup \{a\}$. For every $j \in L \setminus \{i\}$, the bad-incidence condition gives $S \setminus \{j\} = (L \setminus \{i, j\}) \cup \{i, a\} \in G$. These are $r-1$ edges of $G[S]$, while $S \setminus \{a\} = L \notin G$. If $S \setminus \{i\}$ were also an edge, then $G[S]$ would contain r edges and hence a copy of $K_{r+1}^{(r)-}$, a contradiction. Therefore $S \in \mathcal{T}$, and its two missing facets are precisely $S \setminus \{a\}$ and $S \setminus \{i\}$.

Conversely, let $S \in \mathcal{T}$, and write its two missing facets as $S \setminus \{a\}$ and $S \setminus \{b\}$, where $a \neq b$. If $\rho_b = a$, then $(S \setminus \{a\}, b)$ is bad: for every $j \in S \setminus \{a, b\}$, we have $(S \setminus \{a, b, j\}) \cup \{b, a\} = S \setminus \{j\} \in G$. Similarly, if $\rho_a = b$, then $(S \setminus \{b\}, a)$ is bad. The first part of the proof shows that these are the only bad incidences associated with S . Each of the two root events has probability $1/(n-1)$. By linearity of expectation, every $S \in \mathcal{T}$ contributes exactly $2/(n-1)$ to the expected number of bad incidences. Summing over $\mathcal{T}$ proves the first identity, and the second follows from the definition of w .

The Johnson packing and the repair.

For every realization of the roots, apply [Lemma 2.3](#) to the weights w on $\mathcal{C}$. We obtain an independent set $D \subseteq \mathcal{C}$, containing only positive-weight vertices, such that $\sum_{L \in D} w(L) \geq n^{-1} \sum_{L \in \mathcal{C}} w(L)$. In particular, every $L \in D$ satisfies $c(L) \leq 1$. Moreover, the independence of D implies that every $(r+1)$ -subset of $[n]$ contains at most one member of D .

The choice of D may depend on the roots, but the preceding packing inequality holds pointwise for every root realization. Combining it with [Equation \(3.1\)](#) and the bad-incidence identity above, and then averaging over the roots, shows that some realization and its corresponding set D satisfy

$$\frac{1}{2} \sum_{i=1}^n |B_i| + \sum_{L \in D} w(L) \geq \frac{s}{n-1} |G| + \frac{1}{n} \left(|\mathcal{C}| - \frac{|\mathcal{T}|}{n-1} \right).$$

Fix such a realization and such a set D .

For every $L \in D$ with $c(L) = 1$, let j be its unique bad label and mark one member of $B_j \cap \binom{L \setminus \{j\}}{r-2}$ for deletion. Such a member exists because the incidence is bad. After all choices have been made, let B'_i be what remains of B_i for each $i \in [n]$. Repeated choices of the same indexed pair-link edge (i, U) can only reduce the number of distinct deletions, and therefore

$$|G| + |D| + \frac{1}{2} \sum_{i=1}^n |B'_i| \geq |G| + |D| + \frac{1}{2} \sum_{i=1}^n |B_i| - \frac{1}{2} \sum_{L \in D} c(L) = |G| + \frac{1}{2} \sum_{i=1}^n |B_i| + \sum_{L \in D} w(L).$$

Together with the preceding lower bound, this proves part (iii).

It remains to verify part (ii). If $L \in D$, the required noncontainment holds for every label when $c(L) = 0$. When $c(L) = 1$, it already holds at all but the unique bad label and is enforced there by the deletion above.

Now let $L \in G$ and fix $i \in L$. Since $B'_i \subseteq B_i$, it is enough to prove the noncontainment for B_i . If $\rho_i \in L$, then at least one member of $\binom{L \setminus \{i\}}{r-2}$ contains ρ_i , while every member of B_i avoids ρ_i . Hence the containment fails.

Suppose that $\rho_i \notin L$ and, for a contradiction, that $\binom{L \setminus \{i\}}{r-2} \subseteq B_i$. For every $j \in L \setminus \{i\}$, we then have $(L \setminus \{j\}) \cup \{\rho_i\} = (L \setminus \{i, j\}) \cup \{i, \rho_i\} \in G$. Together with $L \in G$, these are r edges on the $(r+1)$ -set $L \cup \{\rho_i\}$, contradicting the fact that G is $K_{r+1}^{(r)-}$ -free. This proves compatibility and completes the proof. $\square$

We next convert the repaired packing into a simplex-free construction by a second-order blowup. Fix $G, D, B'_1, \dots, B'_n$ given by [Proposition 3.2](#). For a positive integer m , let $V_1, \dots, V_n$ be pairwise disjoint sets, each of size m , and call i the *label* of each vertex in V_i . We define an r -graph $\mathcal{H}_m$ on $V_1 \cup \dots \cup V_n$. Only the following two label patterns are allowed.

- *Transverse edges.* If an r -set e has one vertex in each of r distinct classes and its label set is $L \in \binom{[n]}{r}$, then $e \in \mathcal{H}_m$ exactly when $L \in G \cup D$.
- *One repeated label.* Suppose that an r -set e has two vertices in V_i , while its remaining $r - 2$ vertices lie in distinct classes with label set U . Then $e \in \mathcal{H}_m$ exactly when $U \in B'_i$.

Every other r -set is a nonedge.

Lemma 3.3. *For every positive integer m , the r -graph $\mathcal{H}_m$ is $K_{r+1}^{(r)-}$ -free.*

Proof. Suppose for a contradiction that an $(r + 1)$ -set $X \subseteq V(\mathcal{H}_m)$ spans a copy of $K_{r+1}^{(r)-}$.

First suppose that X meets $r + 1$ classes. All its vertices have distinct labels. Since G is $K_{r+1}^{(r)-}$ -free, at most $r - 1$ facets of X have label sets in G . Since D is independent in the Johnson graph, at most one facet has its label set in D . Thus at most r of the $r + 1$ facets of X are edges of $\mathcal{H}_m$, a contradiction.

Next suppose that X meets r classes. The multiplicity pattern is $2, 1, \dots, 1$. Let i be the repeated label, and let $L \in \binom{[n]}{r}$ be the set of labels used by X . Deleting either vertex with label i gives a transverse facet, so completeness implies that $L \in G \cup D$. For each $j \in L \setminus \{i\}$, deleting the unique vertex with label j gives a facet with repeated label i and remaining label set $L \setminus \{i, j\}$. Thus $\binom{L \setminus \{i\}}{r-2} \subseteq B'_i$, contrary to part (ii) of [Proposition 3.2](#).

Now suppose that X meets $r - 1$ classes. Since $r \geq 4$, at least one occupied class contains exactly one vertex of X ; otherwise $|X| \geq 2(r - 1) > r + 1$. Delete the vertex in a singleton class. The remaining r -set occupies exactly $r - 2$ classes and therefore has neither of the two allowed label patterns. It is a nonedge, again a contradiction.

Finally, if X meets at most $r - 2$ classes, then every r -subset of X occupies at most $r - 2$ classes and is a nonedge by definition. This covers all cases and proves the lemma. $\square$

Corollary 3.4. *Let $r \geq 4$ and $n > r$, and let G be a $K_{r+1}^{(r)-}$ -free r -graph on $[n]$. With $\mathcal{C}$ and $\mathcal{T}$ as defined before [Proposition 3.2](#), we have*

$$p_{r,r+1} \geq \frac{r!}{n^r} \left[\left(1 + \frac{\binom{r}{2}}{n-1} \right) |G| + \frac{1}{n} \left(|\mathcal{C}| - \frac{|\mathcal{T}|}{n-1} \right) \right].$$

Proof. Choose $D, B'_1, \dots, B'_n$ as in [Proposition 3.2](#), and form $\mathcal{H}_m$ by the preceding second-order blowup construction. Every member of $G \cup D$ contributes exactly m^r transverse edges, while every member of B'_i contributes $\binom{m}{2} m^{r-2}$ edges with repeated label i . Thus $|\mathcal{H}_m| = (|G| + |D|)m^r + (\sum_i |B'_i|) \binom{m}{2} m^{r-2}$. By [Lemma 3.3](#), $\text{ex}(nm, K_{r+1}^{(r)-}) \geq |\mathcal{H}_m|$. Since $\text{ex}(N, K_{r+1}^{(r)-}) / \binom{N}{r} \rightarrow p_{r,r+1}$ as $N \rightarrow \infty$, dividing by $\binom{nm}{r}$ and letting $m \rightarrow \infty$ gives $p_{r,r+1} \geq (r!/n^r)(|G| + |D| + \frac{1}{2} \sum_i |B'_i|)$. The result now follows from [Proposition 3.2](#). $\square$

We now calibrate the finite template estimate at the exact density $p_{r,r}$.

Proof of Theorem 3.1. For $r = 2$, every $K_3^{(2)-}$ -free graph is a matching, so $p_{2,2} = 0$, whereas Mantel's theorem [\[21\]](#) gives $p_{2,3} = 1/2$. For $r = 3$, Markström and Thomassen [\[23, Corollary 3.9\]](#) give $p_{3,3} \leq 1/3$, whereas Turán's construction [\[29\]](#) gives $p_{3,4} \geq 5/9$. Thus $p_{3,4} - p_{3,3} \geq 2/9$.

It remains to consider $r \geq 4$. Fix such an r and an integer $n \geq r + 1$, and put $N := \binom{n}{r}$. Retain $s, A_{r,n}, B_{r,n}$ from the notation preceding [Lemma 2.4](#); in particular, $A_{r,n} = r!N/n^r$. For this proof, set $u := (r - 2)/r$, $\gamma := 1/(r - 1)$, and $K := u\kappa_r + s\gamma$.

Let G be a $K_{r+1}^{(r)-}$ -free r -graph on $[n]$, write $x := |G|/N$, and let $\mathcal{C}, \mathcal{T}$ be as in [Corollary 3.4](#). Every $S \in \mathcal{T}$ contains $r - 1$ edges of G , whereas every edge of G lies in $n - r$ sets of size $r + 1$; hence $(r - 1)|\mathcal{T}| \leq (n - r)|G|$.

Since $|\mathcal{C}| = N - |G|$, [Corollary 3.4](#) gives

$$p_{r,r+1} \geq B_{r,n}x + \frac{A_{r,n}}{n}(1 - \beta_n x), \quad \beta_n := 1 + \frac{n-r}{(r-1)(n-1)}. \quad (3.2)$$

We next calibrate [Equation \(3.2\)](#) to the exact density $p_{r,r}$. By [Lemma 2.1](#), there is a $K_{r+1}^{(r)-}$ -free r -graph on $[n]$ with at least $p_{r,r}N$ edges. Deleting edges produces two such graphs G^- and G^+ with $|G^-| = \lfloor p_{r,r}N \rfloor$ and $|G^+| = \lceil p_{r,r}N \rceil$; if $p_{r,r}N$ is an integer, take $G^- = G^+$. Choose $\theta \in [0, 1]$ so that $(1-\theta)|G^-| + \theta|G^+| = p_{r,r}N$. The right-hand side of [Equation \(3.2\)](#) is affine in x . Taking the same convex combination of the two resulting inequalities and subtracting $p_{r,r}$ yields

$$p_{r,r+1} - p_{r,r} \geq (B_{r,n} - 1)p_{r,r} + \frac{A_{r,n}}{n}(1 - \beta_n p_{r,r}). \quad (3.3)$$

The coefficient of $p_{r,r}$ on the right-hand side of [Equation \(3.3\)](#) is $B_{r,n} - 1 - A_{r,n}\beta_n/n < 0$. The general bound of Markström and Thomassen [[23](#), Corollary 3.9] gives $p_{r,r} \leq u$. Since the coefficient is negative, substituting this upper bound into [Equation \(3.3\)](#) yields $p_{r,r+1} - p_{r,r} \geq u(B_{r,n} - 1) + (A_{r,n}/n)(1 - u\beta_n)$. Moreover, $\beta_n \leq r/(r-1)$ and $1 - u\beta_n \geq 1/(r-1) = \gamma$. If n satisfies the threshold in [Lemma 2.4](#), then that lemma and the last bound give $A_{r,n}(1 - u\beta_n) \geq (1 - s/n)\gamma$. Indeed, if $1 - s/n \geq 0$, this follows by multiplying the two nonnegative lower bounds. If $1 - s/n < 0$, the right-hand side is negative, whereas the left-hand side is positive. Together with $B_{r,n} \geq 1 - \kappa_r/n^2$, this gives

$$p_{r,r+1} - p_{r,r} \geq -\frac{u\kappa_r}{n^2} + \frac{1 - s/n}{n}\gamma = \frac{\gamma}{n} - \frac{K}{n^2}.$$

Now take $n = n_r^{\text{end}}$. Directly from the definitions, $n_r^{\text{end}} = \lceil 2K/\gamma \rceil \geq r(r-1) > r$; this also implies the threshold in [Lemma 2.4](#). Moreover, $0 \leq n_r^{\text{end}} - 2K/\gamma < 1$. Applying [Equation \(2.1\)](#) gives

$$p_{r,r+1} - p_{r,r} \geq \frac{\gamma^2}{4K} \left[1 - \left(\frac{n_r^{\text{end}} - 2K/\gamma}{n_r^{\text{end}}} \right)^2 \right] \geq \frac{\gamma^2}{4K} \left(1 - \frac{1}{(n_r^{\text{end}})^2} \right) = \Delta_r^{\text{end}}.$$

The last equality follows by substituting the definitions of u, γ, K , and κ_r . This proves the $r \geq 4$ assertion of [Theorem 3.1](#). Finally, $\gamma \sim r^{-1}$, $K \sim r^4/8$, and $n_r^{\text{end}} \rightarrow \infty$, proving its asymptotic assertion. $\square$

4 Interior adjacent gaps

Part (i) of [Theorem 1.2](#) follows from a stronger finite estimate. For integers $r \geq 5$ and $4 \leq t \leq r-1$, define

$$n_{r,t} := \left\lceil r(r-1) + \frac{(t-1)(t-2)r(r-1)(r-2)(3r+7)}{12(r-t+2)} \right\rceil.$$

We will in fact prove the following explicit bound.

Theorem 4.1. *For every $r \geq 5$ and $4 \leq t \leq r-1$,*

$$p_{r,t+1} - p_{r,t} \geq \frac{6(r-t+2)^2}{r^2(t-1)(r-1)[(t-1)(t-2)(r-2)(3r+7) + 12(r-t+2)]} \left(1 - \frac{1}{n_{r,t}^2} \right) > 0. \quad (4.1)$$

Correlated hub roots first give two exact expectation identities. A modular Johnson packing and one-edge repair then produce a compatible blowup, and an affine interpolation calibrates the resulting finite inequality at the unknown density $p_{r,t}$. Fix $r \geq 5$ and $4 \leq t \leq r-1$, and write $d := t-1$ and $a := r-t+2$; we retain $s = \binom{r}{2}$ from [Section 2](#). Thus $d \geq 3$ and $3 \leq a \leq r-2$. An $H_t^{(r)}$ -free r -graph has at most d edges on every $(r+1)$ -set, and a set attaining this upper bound has exactly a missing facets.

We begin by defining the pair-links and positive weights used in the hub-root calculation.

Let $n > r$, let $G \subseteq \binom{[n]}{r}$ be $H_t^{(r)}$ -free, and put $\mathcal{C} := \binom{[n]}{r} \setminus G$. Let $\mathcal{T}_t(G)$ be the family of sets $S \in \binom{[n]}{r+1}$ with $|G[S]| = d$; we call these sets *saturated*. Choose a root $\rho_i \neq i$ for every $i \in [n]$. The pair-link B_i consists of the sets $U \in \binom{[n] \setminus \{i, \rho_i\}}{r-2}$ for which $U \cup \{i, \rho_i\} \in G$. For $L \in \binom{[n]}{r}$ and $i \in L$, let $d_i(L) := |B_i \cap \binom{L \setminus \{i\}}{r-2}|$. For $L \in \mathcal{C}$, call (L, i) *bad* if $d_i(L) = d$, let $c(L)$ be the number of bad labels in L ,

and put $w_+(L) := \max\{1 - c(L)/2, 0\}$. Thus $w_+(L)$ records the gain from adding L after charging one half for each required pair-link repair; candidates with at least two repair sites receive weight zero.

We use a correlated root distribution. Choose a *hub* h uniformly from $[n]$, then choose b uniformly from $[n] \setminus \{h\}$. Set $\rho_i = h$ for $i \neq h$ and $\rho_h = b$.

Lemma 4.2. *For the pair-links, weights, and hub-root distribution defined above, the following identities hold.*

$$(i) \quad \mathbb{E}_{h,b} \sum_{i=1}^n |B_i| = \frac{r(r-1)}{n-1} |G|.$$

$$(ii) \quad \mathbb{E}_{h,b} \sum_{L \in \mathcal{C}} w_+(L) = |\mathcal{C}| - \left(\frac{a}{n} + \frac{\binom{a}{2}}{n(n-1)} \right) |\mathcal{T}_t(G)|.$$

Proof. For fixed h and b , the sum $\sum_{i \neq h} |B_i|$ counts every edge through h exactly $r-1$ times, whereas $|B_h|$ is the codegree of $\{h, b\}$. Averaging the first quantity over h gives $r(r-1)|G|/n$, and averaging the second over the ordered pairs (h, b) gives $r(r-1)|G|/(n(n-1))$. Their sum proves part (i).

For part (ii), fix $L \in \mathcal{C}$. We separate two cases.

Suppose first that $h \notin L$ and put $S := L \cup \{h\}$. Every $i \in L$ has root h . For $x \in L \setminus \{i\}$, the set $L \setminus \{i, x\}$ belongs to B_i if and only if the facet $S \setminus \{x\}$ belongs to G . Thus $d_i(L)$ counts exactly these facets. If (L, i) is bad, the d counted facets are edges of $G[S]$. The facet $S \setminus \{h\} = L$ is a nonedge, and $S \setminus \{i\}$ must also be a nonedge, since otherwise S would contain at least $d+1 = t$ edges. It follows that $S \in \mathcal{T}_t(G)$ and that h and i index missing facets. Conversely, if $S \in \mathcal{T}_t(G)$, $L = S \setminus \{h\}$ is a missing facet, and $i \neq h$ indexes another missing facet, then the same correspondence shows that the remaining d facets of S are counted by $d_i(L)$, so (L, i) is bad. Hence either $c(L) = 0$, or S is saturated and $c(L) = a-1$. In the saturated case, $c(L) = a-1 \geq 2$, so $w_+(L) = 0$. Each $S \in \mathcal{T}_t(G)$ has a choices for the missing facet $S \setminus \{h\}$, so this case contributes an expected loss of $a|\mathcal{T}_t(G)|/n$.

Now suppose that $h \in L$. If $i \in L \setminus \{h\}$, then $\rho_i = h \in L$, and at most one member of $\binom{L \setminus \{i\}}{r-2}$ can avoid the root. Thus $d_i(L) \leq 1 < d$, and only $i = h$ can be bad. For $b \notin L$, the analogous correspondence sends $L \setminus \{h, x\}$ to the facet $(L \cup \{b\}) \setminus \{x\}$. It follows that (L, h) is bad precisely when $b \notin L$, $S := L \cup \{b\}$ belongs to $\mathcal{T}_t(G)$, and both $S \setminus \{b\}$ and $S \setminus \{h\}$ are missing facets. Every $S \in \mathcal{T}_t(G)$ supplies $a(a-1)$ ordered missing pairs (h, b) , each with probability $1/(n(n-1))$. Such a set L has $c(L) = 1$ and hence loses weight $1/2$. The expected loss in this case is therefore $a(a-1)|\mathcal{T}_t(G)|/(2n(n-1)) = \binom{a}{2}|\mathcal{T}_t(G)|/(n(n-1))$. Subtracting the two losses from the initial weight $|\mathcal{C}|$ proves part (ii). $\square$

We now combine the preceding expectation identities with a modular Johnson packing and local repair, and then form the corresponding second-order blowup.

For each realization of the hub roots, apply Lemma 2.3 to the weights w_+ on $\mathcal{C}$. This gives a Johnson-independent family $D \subseteq \mathcal{C}$ containing only positive-weight members. Thus every $L \in D$ satisfies $c(L) \leq 1$, and $\sum_{L \in D} w_+(L) \geq n^{-1} \sum_{L \in \mathcal{C}} w_+(L)$. For each $L \in D$ with $c(L) = 1$, let j be its unique bad label and mark one member of $B_j \cap \binom{L \setminus \{j\}}{r-2}$ for deletion. After all choices have been made, let B'_i be what remains of B_i for each $i \in [n]$.

Repeated choices of the same indexed pair-link edge can only reduce the number of distinct deletions. Since every $L \in D$ satisfies $c(L) \leq 1$, we also have $w_+(L) = 1 - c(L)/2$. Therefore

$$|D| + \frac{1}{2} \sum_{i=1}^n |B'_i| \geq |D| + \frac{1}{2} \sum_{i=1}^n |B_i| - \frac{1}{2} \sum_{L \in D} c(L) = \frac{1}{2} \sum_{i=1}^n |B_i| + \sum_{L \in D} w_+(L).$$

The repair ensures that no transverse label set can be completed by d facets from one pair-link layer; we verify this compatibility within the blowup argument below.

Let $V_1, \dots, V_n$ be disjoint classes of size m . Define an r -graph $\mathcal{H}_m$ as follows:

- *Transverse edges.* Every transverse r -set whose label set belongs to $G \cup D$ is an edge.
- *One repeated label.* An r -set with two vertices in V_i and one vertex in each of the $r-2$ classes indexed by U is an edge exactly when $U \in B'_i$.

Every other r -set is a nonedge.

Lemma 4.3. *For every positive integer m , the r -graph $\mathcal{H}_m$ is $H_{t+1}^{(r)}$ -free.*

Proof. Compatibility claim. For every $L \in G \cup D$ and every $i \in L$, we have $|B'_i \cap \binom{L \setminus \{i\}}{r-2}| \leq d-1 = t-2$.

First let $L \in G$. If $\rho_i \in L$, then at most one member of $\binom{L \setminus \{i\}}{r-2}$ avoids ρ_i , so this count is at most $1 \leq d-1$. If $\rho_i \notin L$, every pair-link member counted by $d_i(L)$ gives an edge of $G[L \cup \{\rho_i\}]$. There cannot be d such edges, because together with $L \in G$ they would give $d+1 = t$ edges on $r+1$ vertices. Thus the unpaired count is already at most $d-1$.

Now let $L \in D \subseteq \mathcal{C}$. Before repair, the corresponding count $d_i(L)$ is at most d . If $\rho_i \in L$, then $d_i(L) \leq 1 \leq d$. If $\rho_i \notin L$, all corresponding facets are edges of $G[L \cup \{\rho_i\}]$, which contains at most d edges.

If $d_i(L) = d$, then (L, i) is bad. Since L has positive weight, it has at most one bad label, and the repair deletes one pair-link member counted at that label. Even if another bad incidence chose the same indexed member, that member is still absent from the final B'_i , so the count for every bad incidence decreases by at least one. If $d_i(L) < d$, no deletion is needed. In both cases the repaired count is at most $d-1$.

This proves the compatibility claim.

We now prove the asserted freeness. It suffices to show that every $(r+1)$ -set X spans at most t edges. If X has $r+1$ distinct labels, then G contributes at most $d = t-1$ facets. Since D is Johnson-independent, it contributes at most one more, for a total of at most t .

If X has r distinct labels, its multiplicity pattern is $2, 1, \dots, 1$. Let i be the repeated label and let L be its r -element label set. For each $x \in L \setminus \{i\}$, deleting the vertex with label x produces a facet with repeated label i and remaining label set $L \setminus \{i, x\}$. Thus these facets are counted exactly by $B'_i \cap \binom{L \setminus \{i\}}{r-2}$.

If $L \in G \cup D$, the two transverse facets and the compatibility claim give at most $2 + (d-1) = t$ edges. If $L \notin G \cup D$, there are no transverse facets. When $\rho_i \notin L$, the facets with repeated label i correspond to edges of $G[L \cup \{\rho_i\}]$, so there are at most d ; when $\rho_i \in L$, there is at most one. Again the total is at most t .

If X has $r-1$ distinct labels, the only multiplicity patterns that can produce allowed facets are $3, 1^{r-2}$ and $2, 2, 1^{r-3}$. They produce at most three and four facets with one repeated label, respectively. Hence each produces at most $4 \leq t$ edges. If X has at most $r-2$ distinct labels, no facet has an allowed label pattern. This proves the lemma. $\square$

Proposition 4.4. *Under the standing assumptions $r \geq 5$, $4 \leq t \leq r-1$, and $n > r$, every $H_t^{(r)}$ -free r -graph G on $[n]$ satisfies*

$$p_{r,t+1} \geq \frac{r!}{n^r} \left[\left(1 + \frac{s}{n-1}\right) |G| + \frac{1}{n} |\mathcal{C}| - \frac{1}{n} \left(\frac{a}{n} + \frac{\binom{a}{2}}{n(n-1)} \right) |\mathcal{T}_t(G)| \right].$$

Proof. The two allowed edge types contribute $(|G| + |D|)m^r$ and $(\sum_i |B'_i|) \binom{m}{2} m^{r-2}$ edges, respectively. Dividing their sum by $\binom{nm}{r}$ and letting $m \rightarrow \infty$ gives, by Lemma 4.3,

$$p_{r,t+1} \geq \frac{r!}{n^r} \left(|G| + |D| + \frac{1}{2} \sum_{i=1}^n |B'_i| \right).$$

For every root realization, the packing and repair estimates above show that the expression in parentheses is at least $|G| + \frac{1}{2} \sum_{i=1}^n |B_i| + \frac{1}{n} \sum_{L \in \mathcal{C}} w_+(L)$. Averaging this lower bound over (h, b) and using both parts of Lemma 4.2, we find a realization whose value is at least the right-hand side in the proposition. $\square$

We next combine the finite template with a saturated-incidence bound and interpolate at the limiting density $p_{r,t}$. Retain the product notation $A_{r,n}$ and $B_{r,n}$ from the notation preceding Lemma 2.4.

For every $H_t^{(r)}$ -free r -graph G , each $S \in \mathcal{T}_t(G)$ contains d edges of G , and each edge of G belongs to $n-r$ sets of size $r+1$. Saturated incidences form a subfamily of all edge- $(r+1)$ -set incidences, so $d|\mathcal{T}_t(G)| \leq (n-r)|G|$. For the remainder of this section, put $u := (d-1)/r$, $\gamma := a/(rd)$, and $K := u\kappa_r + s\gamma$.

Also define $\beta_n := 1 + \frac{a(n-r)}{dn} \left(1 + \frac{a-1}{2(n-1)}\right)$. Since $(a-1)(n-r) \leq 2r(n-1)$, the factor $\frac{n-r}{n} \left(1 + \frac{a-1}{2(n-1)}\right)$ is at most 1; hence $\beta_n \leq 1 + a/d = (r+1)/d$. Consequently, $1 - u\beta_n \geq 1 - (d-1)(r+1)/(rd) = a/(rd) = \gamma$.

Proof of Theorem 4.1. Fix an integer $n \geq r+1$ and put $N := \binom{n}{r}$. By Lemma 2.1, there is an $H_t^{(r)}$ -free r -graph on $[n]$ with at least $p_{r,t}N$ edges. Deleting edges from it gives $H_t^{(r)}$ -free subgraphs G^- and G^+ satisfying $|G^-| = \lfloor p_{r,t}N \rfloor$ and $|G^+| = \lceil p_{r,t}N \rceil$; if $p_{r,t}N$ is an integer, take $G^- = G^+$. Choose $\theta \in [0, 1]$ such that $(1-\theta)|G^-| + \theta|G^+| = p_{r,t}N$.

For either $G \in \{G^-, G^+\}$, write $x := |G|/N$. The saturated-incidence bound gives $|\mathcal{T}_t(G)|/N \leq (n-r)x/d$. The saturated-set term in Proposition 4.4 has a negative coefficient. Hence that proposition gives

$$\begin{aligned} p_{r,t+1} &\geq A_{r,n} \left[\left(1 + \frac{s}{n-1}\right) x + \frac{1-x}{n} - \frac{n-r}{dn} \left(\frac{a}{n} + \frac{\binom{a}{2}}{n(n-1)} \right) x \right] \\ &= B_{r,n}x + \frac{A_{r,n}}{n} (1 - \beta_n x). \end{aligned}$$

The right-hand side is affine in x . Since the same left-hand side dominates this bound for both G^- and G^+ , it also dominates their convex combination with coefficients $1-\theta$ and θ . The combined value of x is exactly $p_{r,t}$, and subtracting $p_{r,t}$ gives the affine lower bound

$$p_{r,t+1} - p_{r,t} \geq (B_{r,n} - 1)p_{r,t} + \frac{A_{r,n}}{n} (1 - \beta_n p_{r,t}). \quad (4.2)$$

We now turn Equation (4.2) into a quadratic lower bound. The coefficient of $p_{r,t}$ on its right-hand side is $B_{r,n} - 1 - A_{r,n}\beta_n/n < 0$, because $B_{r,n} \leq 1$ by Lemma 2.4 and $A_{r,n}\beta_n/n > 0$. Markström and Thomassen [23, Corollary 3.9] proved that $p_{r,t} \leq (t-2)/r = (d-1)/r = u$. In their notation, the number of deleted edges is $r+1-t = a-1$, and indeed $1-a/r = (t-2)/r$. Since the coefficient above is negative, we may substitute this unconditional upper bound to obtain $p_{r,t+1} - p_{r,t} \geq u(B_{r,n} - 1) + (A_{r,n}/n)(1 - u\beta_n)$. If n satisfies the threshold in Lemma 2.4, then the product bounds and the preceding reserve estimate give $A_{r,n}(1 - u\beta_n) \geq (1 - s/n)\gamma$. Indeed, if $1 - s/n \geq 0$, this follows by multiplying the two nonnegative lower bounds. If $1 - s/n < 0$, the right-hand side is negative, whereas the left-hand side is positive. Together with $B_{r,n} \geq 1 - \kappa_r/n^2$, this yields

$$p_{r,t+1} - p_{r,t} \geq -\frac{u\kappa_r}{n^2} + \frac{1 - s/n}{n} \gamma = \frac{\gamma}{n} - \frac{K}{n^2}.$$

By the definitions, the explicit integer introduced before Theorem 4.1 satisfies $n_{r,t} = \lceil 2K/\gamma \rceil \geq r(r-1)$. Since $r(r-1) \geq r(r-1)(r-2)/(4(r+2))$, this choice satisfies the threshold in Lemma 2.4. Moreover, $0 \leq n_{r,t} - 2K/\gamma < 1$. Apply Equation (2.1) to the displayed lower bound with $n = n_{r,t}$. The rounding estimate gives $p_{r,t+1} - p_{r,t} \geq (\gamma^2/(4K))(1 - n_{r,t}^{-2}) > 0$. Substituting the definitions of γ , K , and κ_r and simplifying gives the rational expression in Equation (4.1). $\square$

Proof of part (i) of Theorem 1.2. The ratio of the explicit lower bound in Equation (4.1) to $2a^2/(r^5 d^2 (d-1))$ is

$$\frac{3r^3}{(r-1)(r-2)(3r+7)} \left(1 + \frac{12a}{d(d-1)(r-2)(3r+7)}\right)^{-1} \left(1 - \frac{1}{n_{r,t}^2}\right).$$

Throughout $4 \leq t \leq r-1$, we have $d \geq 3$, $a \leq r-2$, and $n_{r,t} \geq r(r-1)$. Uniformly over this range, the first, second, and third factors in the displayed ratio are $1 + O(r^{-1})$, $1 + O(r^{-1})$, and $1 + O(r^{-4})$, respectively. For the second estimate, note that $0 \leq 12a/(d(d-1)(r-2)(3r+7)) \leq 2/(3r+7)$. Their product tends to 1 uniformly. Hence the displayed ratio is at least $1 - o(1)$ uniformly over the full range, proving the first assertion. The three stated regimes follow by substituting, respectively, $d = t-1$ fixed and $a \sim r$; $d \sim \beta r$ and $a \sim (1-\beta)r$; and $d = r-\ell-1$ and $a = \ell+2$. $\square$

5 The critical gap $p_{r,4} - p_{r,3}$

This section proves part (ii) of Theorem 1.2. For $t = 3$, the pair-link blowup from the preceding sections fails on the multiplicity pattern $2, 2, 1^{r-3}$ because the two repeated labels may each create two pair-link

facets, producing four edges where only three are allowed. We therefore use a second isomorphic template and retain only transverse blowup edges. A random relabeling first supplies a reservoir of candidate edges. A conflict graph records exactly the pairs that cannot be added together, Molloy's coloring bound extracts a large compatible subfamily, and a transverse blowup converts the safe augmentation into a density gain.

We first record the only non-elementary graph-coloring input and a convenient consequence. We write $\chi(J)$, $\chi_\ell(J)$, and $\alpha(J)$ for the chromatic number, list chromatic number, and independence number of a graph J , respectively. All logarithms are natural.

Theorem 5.1 ([24, Theorem 2]). *For every integer $q \geq 4$ and every K_q -free graph of maximum degree $\Delta \geq 4$, the list chromatic number is at most $200q\Delta \log \log \Delta / \log \Delta$.*

Lemma 5.2. *Let $q \geq 4$ and $n \geq e^e$. Every K_q -free graph J with $\Delta(J) \leq n$ has an independent set of size at least $|V(J)| \log n / (400qn \log \log n)$.*

Proof. Write $\Delta := \Delta(J)$. If $\Delta \leq \sqrt{n}$, greedy coloring gives $\alpha(J) \geq |V(J)|/(\Delta + 1) \geq |V(J)|/(2\sqrt{n})$. Since $\sqrt{n} \geq \log n$, $\log \log n \geq 1$, and $q \geq 4$, the coefficient in the claimed bound is smaller than $1/(2\sqrt{n})$. Thus the greedy bound is stronger than the claimed bound.

If $\Delta > \sqrt{n}$, then $\Delta \geq 4$. Since $\chi(J) \leq \chi_\ell(J)$, Theorem 5.1, together with $\Delta \leq n$, $\log \Delta > \frac{1}{2} \log n$, and $\log \log \Delta \leq \log \log n$, gives $\chi(J) \leq 400qn \log \log n / \log n$. A largest color class has the required size. $\square$

Proof of part (ii) of Theorem 1.2. Fix $r \geq 3$ and an integer $n \geq \max\{r+1, \lceil e^e \rceil\}$, to be chosen later, and write $N := \binom{n}{r}$. Let $G \subseteq \binom{[n]}{r}$ be an extremal $H_3^{(r)}$ -free r -graph, and put $x := |G|/N$. By Lemma 2.1, $x \geq p_{r,3}$. We also use Sidorenko's bound [28, Corollary 3.2], which states that $p_{r,3} \geq r^{-2} > 0$. Counting incidences between edges and $(r+1)$ -sets gives $(n-r)|G| \leq 2\binom{n}{r+1}$, and hence $x \leq 2/(r+1)$ and $1-x \geq (r-1)/(r+1)$.

For $\sigma \in S_n$, let $G^\sigma := \{\sigma(E) : E \in G\}$. If σ is uniform, then each fixed r -set belongs to G^σ with probability x , and hence $\mathbb{E}_\sigma |G \cap G^\sigma| = x^2 N$. Fix a permutation satisfying $|G \cap G^\sigma| \leq x^2 N$, and define the candidate family $\mathcal{A} := G^\sigma \setminus G$. Then $|\mathcal{A}| \geq x(1-x)N$.

Pairs of candidates need not be safe to add simultaneously. We encode this obstruction in a graph $\Gamma := \Gamma(G, G^\sigma)$ on vertex set $\mathcal{A}$, joining distinct L, L' when $|L \cap L'| = r-1$ and $|G[L \cup L']| = 2$. An edge of Γ is therefore precisely a pair of candidates whose simultaneous insertion would create four edges on one $(r+1)$ -set.

Conflict-graph claim. The graph Γ satisfies $\Delta(\Gamma) \leq n-r < n$ and is $K_{R_{r-1}(3)}$ -free. Indeed, fix $L \in \mathcal{A}$. Every neighbor L' determines a unique vertex $v \in [n] \setminus L$ such that $L \cup L' = L \cup \{v\}$. Since $L \in G^\sigma$ and G^σ is $H_3^{(r)}$ -free, the set $L \cup \{v\}$ contains at most one further edge of G^σ . Hence each v produces at most one neighbor, and there are $n-r$ choices for v .

For the clique bound, let $\mathcal{K}$ be a clique of Γ with at least three members. By the standard clique classification for Johnson graphs [25, Proposition 1], either all members of $\mathcal{K}$ contain one fixed $(r-1)$ -set, or all are contained in one fixed $(r+1)$ -set. The latter alternative is impossible because $\mathcal{K} \subseteq G^\sigma$ and an $(r+1)$ -set contains at most two edges of G^σ . Hence all members of $\mathcal{K}$ contain a fixed $Q \in \binom{[n]}{r-1}$. Define $X_Q := \{a \in [n] \setminus Q : Q \cup \{a\} \in \mathcal{K}\}$, so that $\mathcal{K} = \{Q \cup \{a\} : a \in X_Q\}$.

For distinct $a, b \in X_Q$, put $S_{ab} := Q \cup \{a, b\}$. The candidate facets $Q \cup \{a\}$ and $Q \cup \{b\}$ are not in G , while adjacency in Γ gives $|G[S_{ab}]| = 2$. Thus exactly two vertices $z \in Q$ satisfy $S_{ab} \setminus \{z\} \in G$; choose one as the color $\varphi(ab)$ of $ab \in \binom{X_Q}{2}$. This colors the complete graph on X_Q with at most $r-1$ colors.

If $|X_Q| \geq R_{r-1}(3)$, then there are distinct $a, b, c \in X_Q$ and $z \in Q$ such that $\varphi(ab) = \varphi(ac) = \varphi(bc) = z$. By this identity, the three sets

$$(Q \setminus \{z\}) \cup \{a, b\}, \quad (Q \setminus \{z\}) \cup \{a, c\}, \quad (Q \setminus \{z\}) \cup \{b, c\}$$

are edges of G , all contained in the same $(r+1)$ -set $(Q \setminus \{z\}) \cup \{a, b, c\}$. This contradicts $H_3^{(r)}$ -freeness and proves the claim.

Set $q := R_{r-1}(3) \geq R_2(3) = 6$. The conflict-graph claim and Lemma 5.2 give an independent family $D \subseteq \mathcal{A}$ satisfying $|D| \geq |\mathcal{A}| \log n / (400qn \log \log n)$.

Safe augmentation. Put $F := G \cup D$. We verify directly that F is $H_4^{(r)}$ -free. Let $S \in \binom{[n]}{r+1}$. Since $D \subseteq \mathcal{A} = G^\sigma \setminus G$, the families G and D are edge-disjoint. Moreover, since $D \subseteq G^\sigma$, we have $|D[S]| \leq 2$. If $|G[S]| \leq 1$, then $|F[S]| \leq 3$. If $|G[S]| = 2$ and $|D[S]| = 2$, the two members of $D[S]$ are adjacent in Γ by its definition, a contradiction. Hence $|D[S]| \leq 1$ in the saturated case as well, and therefore F is $H_4^{(r)}$ -free.

Recall that $A_{r,n} = r! \binom{n}{r} / n^r$ is the transverse blowup factor. Since $H_4^{(r)}$ is pair-covering, Sidorenko's blowup lemma [28, Lemma 2.2] gives

$$p_{r,4} \geq \frac{r!}{n^r} |F| = A_{r,n} \left(x + \frac{|D|}{N} \right).$$

We now combine these estimates to obtain the stated quantitative bounds. Recall that $s = \binom{r}{2}$ denotes the first-order loss from the transverse blowup factor. Put $\eta_r := (r-1)/(400R_{r-1}(3)(r+1))$, so that $M_r = s/\eta_r + 1$. Combining the candidate-mass, independent-family, and incidence bounds above gives $|D|/N \geq p_{r,3}\eta_r \log n / (n \log \log n)$. The product bounds give $A_{r,n} \geq 1 - s/n$. Since $x \geq p_{r,3}$ and $A_{r,n} > 0$, we also have $A_{r,n}x - p_{r,3} \geq (A_{r,n} - 1)p_{r,3} \geq -sp_{r,3}/n$. Together with the lower bound on $|D|/N$, this gives

$$p_{r,4} - p_{r,3} \geq \frac{p_{r,3}}{n} \left[A_{r,n}\eta_r \frac{\log n}{\log \log n} - s \right]. \quad (5.1)$$

For fixed r , the bracket is positive for all sufficiently large n . Together with Sidorenko's bound, this already proves $p_{r,3} < p_{r,4}$.

It remains to verify the explicit estimates in part (ii) of Theorem 1.2. The definitions give $M_r = 200R_{r-1}(3)r(r+1) + 1$. Choose $n := M_r^{2M_r}$. This integer is larger than $\max\{r+1, e^e\}$. Since $M_r \geq 3$ and $2 \log M_r \leq M_r$, we have $\log n / \log \log n = 2M_r \log M_r / \log(2M_r \log M_r) \geq M_r$. Moreover, $\eta_r < 1/2$, so $M_r = s/\eta_r + 1 > 2s$. Since $n \geq M_r^2$, it follows that $n \geq 2sM_r$. Therefore

$$A_{r,n}\eta_r \frac{\log n}{\log \log n} - s \geq \left(1 - \frac{s}{n}\right) \eta_r M_r - s = \eta_r - \frac{s\eta_r M_r}{n} \geq \frac{\eta_r}{2}.$$

Using Sidorenko's bound in Equation (5.1) now gives the first estimate in that part.

For the factorial estimate, we first eliminate the Ramsey parameter. Greenwood and Gleason [13, Theorem 6] proved the standard recurrence $R_k(3) \leq k(R_{k-1}(3) - 1) + 2$ for $k \geq 2$; together with $R_1(3) = 3$, it gives $R_k(3) \leq 3k!$. Thus the $K_{R_{r-1}(3)}$ -free graph Γ is also $K_{3(r-1)!}$ -free. Apply Lemma 5.2 with $q = 3(r-1)!$. For this choice of q , the coefficient in the preceding calculation is $1/(1200(r+1)(r-2)!)$, and the associated threshold is $1200s(r+1)(r-2)! + 1 = \widehat{M}_r$. Choosing $n := \widehat{M}_r^{2\widehat{M}_r}$ and repeating the preceding calculation proves the second estimate. $\square$

Proof of Theorem 1.1. Since $H_1^{(r)}$ consists of a single edge, $p_{r,1} = 0$. For every $r \geq 2$, Lemma 2.2 gives $p_{r,2} = 0 < p_{r,3}$. If $r \geq 3$, part (ii) of Theorem 1.2 gives $p_{r,3} < p_{r,4}$. For $r \geq 5$, Theorem 4.1 supplies all steps $p_{r,t} < p_{r,t+1}$ for $4 \leq t \leq r-1$. Finally, Theorem 3.1 gives the endpoint $p_{r,r} < p_{r,r+1}$ for every $r \geq 4$. Thus the middle range is empty when $r = 3, 4$, while the endpoint for $r = 3$ is already the critical step. For $r = 2$, the first two observations give the entire chain. $\square$

This completes the proofs of Theorems 1.1 and 1.2. We conclude the section with an anti-Ramsey consequence of the strict adjacent density gaps.

Corollary 5.3. *For every $r \geq 2$ and $3 \leq t \leq r+1$, there is a constant $c_{r,t} > 0$ with the following property. For all sufficiently large n , every integer k satisfying $1 \leq k \leq c_{r,t}\sqrt{n}$ obeys $\text{ar}(n, (k+1)H_t^{(r)}) = \text{ex}(n, kH_t^{(r)}) + 2$. Any fixed constant satisfying $0 < c_{r,t} < \sqrt{(p_{r,t} - p_{r,t-1})/(2r(r+1)t)}$ is valid. At the endpoint $t = r+1$, for $r \geq 4$ one may take*

$$c_{r,r+1} = \left(1 - \frac{1}{r}\right) \sqrt{\frac{\Delta_r^{\text{end}}}{2r(r+1)^2}} = (1 - o(1))r^{-9/2}.$$

In particular, for every $r \geq 2$ and $3 \leq t \leq r+1$, $\text{ar}(n, 2H_t^{(r)}) = \text{ex}(n, H_t^{(r)}) + 2$ for all sufficiently large n .

Proof. Fix r and t , and put $F := H_t^{(r)}$. Let $\mathcal{U}_{r,t}$ consist of F and all its *edge-sums*. To form an edge-sum, start with two vertex-disjoint copies of F , delete one edge from each, and identify the vertices of the two deleted edges according to a bijection; make no other identifications.

All one-edge deletions of F are isomorphic to $H_{t-1}^{(r)}$; let F^- denote this isomorphism type. Its 2-blowup $F^-[2]$ is obtained by replacing each vertex with two clones and each edge with the corresponding complete r -partite r -graph. In the proof of Proposition 1.6(i), Deng et al. [6] observed that every edge-sum-free r -graph is $F^-[2]$ -free. Keevash's blowup-invariance theorem [17, Theorem 2.2] gives $\pi(\mathcal{U}_{r,t}) \leq \pi(F^-[2]) = \pi(F^-) = p_{r,t-1}$. Consequently, $(\text{ex}(n, F) - \text{ex}(n, \mathcal{U}_{r,t})) / \binom{n}{r}$ tends to $p_{r,t} - \pi(\mathcal{U}_{r,t}) \geq p_{r,t} - p_{r,t-1} > 0$, where the last inequality follows from Theorem 1.1.

Deng et al. [6, Theorem 1.5] proved that the claimed anti-Ramsey equality holds for every positive integer k satisfying

$$k \leq \sqrt{\frac{\text{ex}(n, F) - \text{ex}(n, \mathcal{U}_{r,t})}{2v(F)|F|\binom{n-1}{r-1}}}.$$

Since $v(F) = r+1$, $|F| = t$, and $\binom{n}{r} / \binom{n-1}{r-1} = n/r$, dividing this threshold by $\sqrt{n}$ and letting $n \rightarrow \infty$ gives

$$\sqrt{\frac{p_{r,t} - \pi(\mathcal{U}_{r,t})}{2r(r+1)t}} \geq \sqrt{\frac{p_{r,t} - p_{r,t-1}}{2r(r+1)t}}.$$

Thus every fixed $c_{r,t}$ in the stated range is valid for all sufficiently large n . When $t = r+1$, Theorem 3.1 gives $p_{r,r+1} - p_{r,r} \geq \Delta_r^{\text{end}}$ for $r \geq 4$, so the stated endpoint choice is valid; its asymptotic order follows from $\Delta_r^{\text{end}} = (2 - o(1))r^{-6}$. Taking $k = 1$ proves the last assertion. $\square$

6 Interlacing across uniformities

The preceding sections establish strictness at each fixed uniformity. We first give an ordered deletion lift that does not decrease normalized density when passing from uniformity $r-k$ to r .

Theorem 6.1. *Let $r \geq 3$, $1 \leq k \leq r-2$, $1 \leq t \leq r-k+1$, and $n \geq r+1$. Then*

$$\frac{\text{ex}(n, H_{t+k}^{(r)})}{\binom{n}{r}} \geq \frac{\text{ex}(n, H_t^{(r-k)})}{\binom{n}{r-k}}.$$

Consequently, $p_{r,t+k} \geq p_{r-k,t}$.

Proof. Let $P \subseteq \binom{[n]}{r-k}$ be $H_t^{(r-k)}$ -free, and choose a total order $\prec$ on $[n]$. Let $L_k(P)$ consist of the r -sets for which deleting the k smallest elements leaves a member of P .

Consider $S = \{v_0 \prec v_1 \prec \dots \prec v_r\}$ and put $U := \{v_k, \dots, v_r\}$ and $Q_0 := \{v_{k+1}, \dots, v_r\}$. Because P is $H_t^{(r-k)}$ -free, every $(r-k+1)$ -set spans at most $t-1$ members of P . For $0 \leq i \leq k$, deleting v_i from S and then deleting the k smallest remaining elements leaves Q_0 . For $i > k$, the resulting set is $U \setminus \{v_i\}$. Hence $|L_k(P)[S]| = |P[U]| + k\mathbf{1}_{\{Q_0 \in P\}}$. If $Q_0 \notin P$, the right-hand side is at most $t-1$. If $Q_0 \in P$, it is at most $t+k-1$. Thus $L_k(P)$ is $H_{t+k}^{(r)}$ -free.

Now choose $\prec$ uniformly at random. For a fixed r -set, the set left after deleting its k smallest elements is uniform among its $\binom{r}{k}$ subsets of size $r-k$. Double-counting pairs (E, Q) with $Q \subseteq E$, $|Q| = r-k$, and $Q \in P$ gives $\mathbb{E}_{\prec}[|L_k(P)|] / \binom{n}{r} = |P| / \binom{n}{r-k}$. Some order attains at least this normalized density. Taking P extremal proves the finite assertion, and passing to the limit proves the density assertion. $\square$

The opposite comparison follows by iterating Sidorenko's link monotonicity [28, equation (1)]: whenever $r \geq 3$, $1 \leq k \leq r-2$, and $1 \leq t \leq r-k+1$, it gives $p_{r,t} \leq p_{r-k,t}$.

Corollary 6.2. *Let $r \geq 3$ and $1 \leq k \leq r-2$. For every $1 \leq t \leq r-k+1$,*

$$p_{r,t} \leq p_{r-k,t} \leq p_{r,t+k}.$$

In particular, for $1 \leq t \leq r$, $p_{r,t} \leq p_{r-1,t} \leq p_{r,t+1}$. For every $1 \leq j \leq r-k+1$, the covering densities satisfy $\tau_j^{(r)} \leq \tau_j^{(r-k)} \leq \tau_{j+k}^{(r)}$.

Proof. The first density inequality is the preceding link comparison, while the second is the density conclusion of [Theorem 6.1](#). For the covering statement, apply the density interlacing with $t = r - k + 2 - j$. The complement relation $\tau_j^{(r')} = 1 - p_{r', r'+2-j}$ turns these two inequalities, in reverse order, into the claimed covering interlacing. $\square$

We next use this interlacing to decompose adjacent gaps and control their tail sums. For integers $r \geq t \geq 3$, write $g_{r,t} := p_{r,t+1} - p_{r,t}$. By [Corollary 6.2](#),

$$g_{r,t} = (p_{r-1,t} - p_{r,t}) + (p_{r,t+1} - p_{r-1,t}). \quad (6.1)$$

Both summands are nonnegative.

Proposition 6.3. *Fix $t \geq 3$. The following bounds hold.*

(i) *For every $m \geq t - 1$ the first inequality below holds, and for every $m \geq t$ the second one holds:*

$$\sum_{r=m+1}^{\infty} g_{r,t} \geq p_{m,t} \quad \text{and} \quad \sum_{r=m}^{\infty} g_{r,t} \geq p_{m,t+1}.$$

(ii) *Let $m \geq t$ and suppose that $0 < b_m \leq p_{m,t+1}$. Then, for every integer $R \geq m$,*

$$\sum_{r=m}^R g_{r,t} \geq b_m - \frac{t-2}{R} \quad \text{and} \quad \max_{m \leq r \leq R} g_{r,t} \geq \frac{b_m - (t-2)/R}{R - m + 1}.$$

Moreover, if $m \rightarrow \infty$ through integers and (b_m) is a positive sequence satisfying $b_m \leq p_{m,t+1}$, set $R_m := \lceil 2(t-2)/b_m \rceil$. Then

$$\max_{m \leq r \leq R_m} g_{r,t} \geq \left(\frac{1}{4(t-2)} - o(1) \right) b_m^2.$$

Proof. For part (i), let $R > m$. The first summand in [Equation \(6.1\)](#) gives

$$\sum_{r=m+1}^R g_{r,t} \geq p_{m,t} - p_{R,t}.$$

Since $p_{R,t} \leq (t-2)/R$ by Markström and Thomassen [[23](#), Corollary 3.9], letting $R \rightarrow \infty$ proves the first assertion.

Interlacing also gives $p_{r,t} \leq p_{r+1,t+1} \leq p_{r,t+1}$ and hence $g_{r,t} \geq p_{r,t+1} - p_{r+1,t+1}$. Summing from $r = m$ to R , using again Markström and Thomassen's bound $p_{R+1,t+1} \leq (t-1)/(R+1)$ [[23](#), Corollary 3.9], and then letting $R \rightarrow \infty$ proves the second assertion.

For part (ii), keep $g_{m,t}$ intact and, for each $r > m$, retain only the first nonnegative summand in [Equation \(6.1\)](#). Then

$$\sum_{r=m}^R g_{r,t} \geq g_{m,t} + \sum_{r=m+1}^R (p_{r-1,t} - p_{r,t}) = p_{m,t+1} - p_{R,t} \geq b_m - \frac{t-2}{R}.$$

This proves the first displayed bound; averaging proves the second. For the asymptotic assertion, the general upper bound of Markström and Thomassen [[23](#), Corollary 3.9] gives $b_m \leq (t-1)/m \leq 2(t-2)/m$, so $R_m \geq m$. This choice makes the numerator at least $b_m/2$, while $R_m = 2(t-2)/b_m + O(1)$. Since $R_m - m + 1 \leq R_m$, this proves the asymptotic assertion. $\square$

Finally, we combine these tail estimates with Sidorenko's fixed-level lower bound [[28](#), Corollary 3.2 and equation (2)], which states that, for every fixed $k \geq 3$,

$$p_{r,k} \geq \left(((k-2)!)^{1/(k-2)} \Gamma\left(\frac{k-1}{k-2}\right) - o(1) \right) r^{-(1+\frac{1}{k-2})} \quad (r \rightarrow \infty).$$

For fixed $t \geq 3$, put $\beta_t := t/(t-1)$ and $C_t := ((t-1)!)^{1/(t-1)} \Gamma(\beta_t)$.

Corollary 6.4. *For every fixed $t \geq 3$,*

$$\limsup_{r \rightarrow \infty} r^{\beta_t+1} g_{r,t} \geq \beta_t C_t.$$

Moreover, for every sufficiently large m , there exists an integer r satisfying

$$m \leq r \leq \left(\frac{2(t-2)}{C_t} + o(1) \right) m^{\beta_t} \quad \text{and} \quad g_{r,t} \geq \left(\frac{C_t^2}{4(t-2)} - o(1) \right) m^{-2\beta_t}.$$

Here the $o(1)$ terms tend to zero as $m \rightarrow \infty$, with t fixed.

Proof. Applying Sidorenko's bound above at level $k = t + 1$ gives $p_{m,t+1} \geq (C_t - o(1))m^{-\beta_t}$. Suppose the limsup assertion fails. Then, for some $\varepsilon > 0$ and all sufficiently large r , we have $g_{r,t} \leq (\beta_t C_t - \varepsilon)r^{-\beta_t-1}$. It follows that

$$\sum_{r=m}^{\infty} g_{r,t} \leq (\beta_t C_t - \varepsilon) \sum_{r=m}^{\infty} r^{-\beta_t-1} = \left(C_t - \frac{\varepsilon}{\beta_t} + o(1) \right) m^{-\beta_t},$$

contradicting part (i) of Proposition 6.3 together with the preceding Sidorenko lower bound. This proves the limsup assertion.

For the window statement, choose a positive sequence $b_m \leq p_{m,t+1}$ satisfying $b_m = (C_t - o(1))m^{-\beta_t}$. Then the endpoint in the preceding proposition satisfies $R_m = (2(t-2)/C_t + o(1))m^{\beta_t}$. Applying part (ii) of Proposition 6.3 gives the window assertion. $\square$

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Declaration on the use of AI

The authors used generative AI tools to discuss proof organization, test local calculations, check proofs, and improve exposition.

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